Open problems:
Projective measurements are equilibration processes

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`).

From the brief

1. **Unbounded observables.** For unbounded X the Lindblad equation of EQ-LIND holds only on a domain; the brief claims that the blockwise solution (5) "extends the resulting semigroup to every density operator" and that $\rho(t)$ stays a density operator. Neither the extension nor Theorem 1 for unbounded X is proved. (Brief, Sec. 3, after eq. (5).)

2. **A formalism with finite entropy.** *Research.* When $S(P_X(\rho)) = +\infty$, EQ-KL does not determine a closest ensemble. The brief suggests that "the real problem is the Hilbert space formalism, as it does not guarantee finite entropy for all density operators"; a state space in which every ensemble has finite entropy is not given. (Brief, Sec. 4, "Relative entropy in infinite dimensions".)

3. **Other dynamics with the same limit.** *Research.* "Other dynamics may have the same asymptotic map": which Lindblad (or other) dynamics have $\mathcal{P}_X$ as their long-time limit, and whether physical measurements are equilibrations "that happen at a fast scale", is left open. (Brief, Sec. 4, "Equilibration as a black-box description"; Sec. 5.)

4. **Destructive measurements.** *Research.* "The black-box idea could be extended to destructive measurements", with the system state as input and "the final state of the measurement device" as output. No such characterization is given. (Brief, Sec. 5.)

From the formalization

None found. Beyond finite dimension the formalization has a part in any dimension, which covers bounded observables with pure point spectrum, and the failure of the conditions on a continuous part of the spectrum; it leaves open the item "Unbounded observables" above.